Name_____ Date_____ Period_____

LESSON 6-6 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Simplifying to Show Equivalence

Maryland Standard 6.EE.A.3 · Learning target – I can simplify an expression by combining like terms to test whether two expressions are equivalent.

REMEMBER

- **Like terms** have the same variable part: $4x$ and $9x$ are like terms. $4x$ and 4 are not.
- Combine like terms by adding or subtracting coefficients: $4x + 9x = 13x$. A lone x is $1x$.
- Expressions are **equivalent** if they simplify to the same expression.

WORKED EXAMPLE

Are $5m + 8 + 2m - 3$ and $7m + 5$ equivalent?

Step 1 Combine the m terms: $5m + 2m = 7m$

Step 2 Combine the constants: $8 - 3 = 5$

Step 3 The first simplifies to $7m + 5$, the same as the second.

Answer **Yes, they are equivalent.**

WORD BANK

like terms – terms with the same letter part, like 4x and 9x

coefficient – the number in front of the letter

constant – a number with no letter

simplify – write it in a shorter way

PRACTICE

show your work · label units

1. Simplify each expression.

a. $6a + 3a =$ _____

b. $10y - 4y =$ _____

c. $x + 5x + 2 =$ _____

d. $7 + 3n + 4 =$ _____

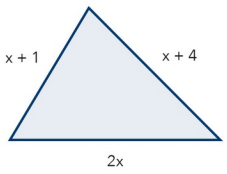
e. $8b + 6 - 3b + 1 =$ _____

f. $2p + 9q + 5p =$ _____

g. $4m + 2 + m + 9 =$ _____

h. $12z - z =$ _____

2. Perimeter. The side lengths of the triangle are in centimeters. Write and simplify an expression for its perimeter.



Step 1 Add all three side lengths.

Step 2 Combine the x terms, then the numbers.

Show your work.

Answer_____ cm

3. Are $3w + 3w$ and $6w^2$ equivalent?

Step 1 Simplify $3w + 3w$.

Step 2 Compare it with $6w^2$.

☐ Yes ☐ No

4. Which expression is equivalent to $5x + 3 + 2x$?

Step 1 Combine $5x$ and $2x$.

Step 2 Keep the 3 by itself. Find the match.

☐ $10x$ ☐ $7x + 3$ ☐ $5x + 5$

5. Find the Error. Elena simplified $7x + 4 - 2x$ and wrote $9x$. Explain her mistake. Then simplify correctly.

Frame *Elena added the ___ to the x terms. The correct answer is ___.*

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Explain to someone at home why $4x + 4$ cannot be written as $8x$.

Name_____ Date_____ Period_____

LESSON 6-6 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Testing and Justifying Equivalence

Maryland Standard 6.EE.A.4 · Learning target – I can test whether two expressions are equivalent by substituting values and by reasoning.

REMEMBER

- Equivalent expressions have the **same value for every** value of the variable.
- **Test** with the same value in both. One value that gives different results proves they are **not** equivalent.
- Matching values do not prove it. **Justify** with properties, like the distributive property.

WORKED EXAMPLE

Are $4(n + 3)$ and $4n + 3$ equivalent?

Step 1 Test $n = 2$: $4(2 + 3) = 20$ and $4 \times 2 + 3 = 11$

Step 2 $20 \neq 11$. Distributing shows $4(n + 3) = 4n + 12$.

Answer **Not equivalent**

WORD BANK

equivalent – same value for every number
test – put the same number into both expressions
counterexample – one number that shows they are not the same

PRACTICE

show your work · label units

1. Evaluate each expression when $x = 5$. Which two might be equivalent?

Step 1 Put 5 in place of x in each one.

Step 2 Look for two answers that match.

a. $3(x + 4) =$ ____ **b.** $3x + 12 =$ ____ **c.** $3x + 4 =$ ____

2. Write *E* if the expressions are equivalent or *N* if they are not.

a. $2(a + 5)$ and $2a + 10$ ___ **b.** $6b + 1$ and $7b$ ___

c. $c + c + c$ and c^3 ___ **d.** $8 + 4m - m$ and $3m + 8$ ___

3. Counterexample. Nate says n^2 and $2n$ are the **same** because both equal 4 when $n = 2$. Find **another** value of n that shows he is **wrong**.

Step 1 Pick a new number, like 3.

Step 2 Find n^2 and $2n$ for that number. Compare.

Show your work.

Value and results_____

4. Reason. $3x$ and $x + 6$ are both 9 when $x = 3$. Which value of x shows they are **not** equivalent?

Frame *When $x =$ ___, $3x =$ ___ and $x + 6 =$ ___, so they are not equivalent.*

☐ $x = 3$ ☐ $x = 4$

5. Which property shows that $5(m + 2) = 5m + 10$?

☐ Commutative ☐ Associative ☐ Distributive

TEST PRACTICE

choose the best answer

6. Which expression is **the same as** $7(p + 3)$? Multiply **both** p and 3 by 7.

A $7p + 3$ **B** $7p + 21$ **C** $p + 21$ **D** $10p$

7. Which expression is **not** equivalent to $4y + 8$?

A $4(y + 2)$ **B** $2(2y + 4)$ **C** $8 + 4y$ **D** $4(y + 8)$

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Pick a number with someone at home. Do $2(x + 3)$ and $2x + 6$ give the same value? Try another number.